



Review

Identifying the neurobiology of altered reinforcement sensitivity in ADHD: A review and research agenda

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ABSTRACT

ADHD is associated with altered reinforcement sensitivity, despite a number of inconsistent findings. This review focuses on the overlap and differences between seven neurobiologically valid models and lists 15 predictions assessing reinforcement sensitivity in ADHD. When comparing the models it becomes clear that there are great differences in the level of explanation. For example, some models try to explain a single core deficit in terms lower-level reinforcement systems, such as the dopamine transfer to reward back in time. Other models explain multiple deficits, by describing higher-level systems, such as impaired bottom-up prefrontal activation. When reviewing the available experimental evidence in support of the predictions, most experimental studies have been focusing on behavioral changes in the face of reward and response cost over no-reward, and on delay discounting. There is currently a lack in studies that focus on explaining underlying cognitive or neural mechanisms of altered reinforcement sensitivity in ADHD. Additionally, there is a lack in studies that try to understand what subgroup of children with ADHD shows alterations in reinforcement sensitivity. The scarcity in studies testing the neurobiological predictions is explained partly by a lack in knowledge how to test some of these predictions in humans. Nevertheless, we believe that these predictions can serve as a useful guide to the systematic evaluation of altered reinforcement sensitivity in ADHD.

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1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is one of the most frequently diagnosed childhood disorders and is characterized by inattention, hyperactivity and impulsivity (American Psychiatric Association, 2000). The biological basis of ADHD is still largely unknown, but there is emerging evidence that cognitive impairments in ADHD are subserved by both structural and functional brain abnormalities (Bush et al., 2005; Paloyelis et al., 2007; Seidman et al., 2005). For many years the focus of cognitive research has been on deficits in executive function (e.g., Barkley, 1997; Pennington and Ozonoff, 1996), especially inhibition. Recently, however, an increasing number of theoretical frameworks have incorporated altered reinforcement sensitivity as an important etiological factor (e.g., Barkley, 1997; Blum et al., 2000; Casey et al., 2007; Castellanos and Tannock, 2002; Douglas, 1989; Frank et al., 2007b; Haenlein and Caul, 1987; Newman and Wallace, 1993; Nigg and Casey, 2005; Quay, 1988; Sagvolden et al., 2005; Sergeant et al., 1999; Sonuga-Barke, 2002; Tripp and Wickens, 2008). The current paper focuses on the overlap and differences between seven theories incorporating aspects of altered reinforcement sensitivity. Predictions made or implied by these theories are discussed in terms of future experimental studies.

Evidence for abnormal reinforcement sensitivity in ADHD comes from research at different levels of analyses. At the behavioral level, the positive effect of reinforcement on cognitive skills is larger for children with ADHD than typically developing children (see Luman et al., 2005 for a review). In addition, individuals with ADHD often show a relatively strong preference for options that are rewarding now, but may be unfavorable in the long term (e.g., Drechsler et al., 2008) and they favor small immediate over larger delayed rewards (Sonuga-Barke et al., 1992). At a psychophysiological level, feedback-related autonomic responses (heart rate and skin conductance) of children with ADHD appear to 'normalize' when reward is added to feedback (e.g., Luman et al., 2007), consistent with the improvements in performance which occur when reinforcement is present. At the functional neuroimaging level, children with ADHD demonstrate reduced activity in the ventral striatum when anticipating reward (e.g., Scheres et al., 2007) which offers a possible explanation for the stronger preference for reward immediacy in ADHD compared to typically developing children. In addition, children with ADHD display reduced positive event-related potential (ERP) activity 200 ms following monetary losses (e.g., van Meel et al., 2005), indicative of a compromised categorization of motivationally relevant stimuli.

An increasing number of experimental studies on reinforcement sensitivity are appearing in the literature. However, the findings are not entirely consistent, confirming the complexity of altered reinforcement sensitivity in ADHD. For example, only some studies report disproportional improvement in performance in response to reward in ADHD groups (see Luman et al., 2005). Even the most consistent finding, namely that children with ADHD show a relatively strong preference for small immediate rewards over larger delayed rewards (Sonuga-Barke et al., 1992; Rapport et al., 1986), may depend on a number of contextual factors (Scheres

et al., 2006, in press). Besides the heterogeneity in the experimental findings, there seems to be a large gap between the experimental findings on the one hand and the theoretical models on the other. An analysis of five theoretical frameworks of ADHD (Douglas, 1989; Haenlein and Caul, 1987; Quay, 1988; Sergeant et al., 1999; Sonuga-Barke, 2002) showed that none of the frameworks were able to explain the experimental findings of the 21 experimental studies evaluated (Luman et al., 2005).

Luman et al. suggested that the inability of the theoretical frameworks to account for the experimental results might be due to the domain specificity of the different frameworks. For example, although the frameworks as offered by Haenlein and Caul (1987), Douglas (1989) or Sergeant et al. (1999) have been very influential in explaining reinforcement sensitivity in ADHD from a behavioral point of view, they do not offer predictions about the underlying (neurobiological) mechanisms of these behaviors. Another important issue contributing to the gap between experimental findings and theory is that current models offer relatively few testable experimental predictions. We believe if the models offered theoretically driven, and experimentally testable, predictions, researchers would be encouraged to conduct the studies necessary to test them. An additional issue is the scarcity of systematic studies 'multi-level studies' into the (neurobiological) mechanisms of reinforcement sensitivity in ADHD. We believe that a multi-level methodological approach, for example by combining behavioral and neuroimaging studies, will lead to increased understanding of the nature of reinforcement sensitivity in ADHD. Up to now, most studies have focused on behavioral outcomes, with a few exceptions (Plichta et al., 2009; Rubia et al., 2009; Scheres et al., 2007; Ströhle et al., 2008; van Meel et al., 2005).

The goal of the current paper is fourfold. (1) To provide an updated review of current theoretical models of reinforcement sensitivity in ADHD; (2) to identify the predictions regarding behavioral or neurobiological responses to reinforcement stimuli, either made by the model developers, or extracted from the models by the current authors; (3) to identify current research methods that are required to test these predictions; (4) to review the existing experimental evidence in support for these predictions. The key elements of these models and their predictions are summarized in Section 2. This helps identify which experimental methods are needed to test the models, which predictions are unique or shared among the models, and which predictions are supported by the available evidence. The predictions can serve as a useful guide to the systematic evaluation of altered reinforcement sensitivity in ADHD, leading to increased understanding of the phenomenon and its etiology.

2. Neuropathological models of altered sensitivity to reinforcement in ADHD

Seven models were selected for inclusion in this review, based on three important criteria. (1) The models must account for some of the behaviors associated with altered reinforcement sensitivity in ADHD; (2) the models should be neurobiologically meaningful. ADHD is assumed to have a biological basis (e.g., Castellanos and Tannock, 2002), and there is good evidence that reinforcement sensitivity can be explained by neurobiological processes (e.g.,

Cools et al., 2007; Knutson et al., 2001, 2003; Schultz, 2000; Schultz et al., 1997); (3) the models must specify experimentally testable behavioral or neurobiological predictions, or such predictions can be logically derived from the models.

Seven models were selected. Two of the models (Sagvolden et al., 2005; Tripp and Wickens, 2008) can be considered as neurochemical deficit models and discuss altered reinforcement sensitivity in ADHD in relation to neurotransmitter functioning. Two models (Nigg and Casey, 2005; Sonuga-Barke, 2002, 2003) can be considered brain-pathway deficit models and discuss ADHD in relation to neuroanatomical pathway functioning. A further two models approach ADHD from a neurocomputational perspective (Frank, 2005; Frank et al., 2007b; Williams and Dayan, 2005). A final model (Newman and Wallace, 1993; Patterson and Newman, 1993) makes specific predictions regarding sensitivity to punishment in ADHD which is not considered by the other models. This model is included here for that reason, despite its lack of an explicit neurobiological basis.

2.1. Neurochemical deficit models

The dynamic developmental theory (DDT; Sagvolden et al., 2005) and the dopamine transfer deficit (DTD) theory (Tripp and Wickens, 2008) are based on the assumption that ADHD is associated with a dysfunction of the midbrain dopamine (DA) system. The animal work by Schultz et al. (1997) demonstrated that the firing rate of DA neurons in the ventral tegmental area (VTA) and substantia nigra increases when an unpredicted reward occurs and decreases when a predicted reward is omitted or in response to aversive stimuli (Schultz et al., 1997; Ungless et al., 2004; Wise, 2004). It is assumed that similar changes occur in humans: Functional neuroimaging studies show an increase in blood-oxygen-level dependent (BOLD) activity that may reflect DA activity in the mesolimbic reward circuitry, including ventral striatum during reward anticipation (Knutson et al., 2001, 2003). Additionally, researchers have shown that BOLD responses in the VTA reflect positive reward prediction errors (D'Ardenne et al., 2008; Murray et al., 2008).

The DDT by Sagvolden et al. (2005) hypothesizes that a dysfunction in DA transmission in the fronto-limbic brain circuitry is responsible for a steeper delay-of-reinforcement gradient and slower effects of extinction (Sagvolden et al., 2005; Johansen et al., 2002). This model proposes that a steeper and shorter delay-of-reinforcement gradient (see Fig. 1b) in children with ADHD is caused by lower levels of tonic DA (see Fig. 1a). A steep gradient indicates that a reinforcer loses its value relatively quickly when the delay between the desired behavior and the reinforcer increases, resulting in impulsive behavior. Accordingly, children with ADHD would prefer immediate over delayed rewards and only show learning when rewards are received immediately and frequently. Further, a steepened delay gradient means that only short sequences of responses can be reinforced in those with ADHD, producing more variable behavior. Since lower tonic DA levels (see Fig. 1a) cause blunted dips in phasic DA after the omission of reward (e.g., Schultz et al., 1997), children with ADHD would show weaker (slower) extinction of behavior.

Tripp and Wickens (2008) DTD theory differs from DDT, in that it assumes altered activity of the phasic DA response to reinforcement, as opposed to low levels of tonic DA, to be the cause of altered reinforcement sensitivity in ADHD. This theory focuses on altered anticipatory firing of DA cells to cues that predict reinforcement (see Fig. 2). The model assumes that in healthy individuals, when an unconditioned stimulus (the reward) is repeatedly coupled with a conditioned stimulus (the predictor), DA neurons begin firing to the predictor as well as the reward. Once the reward-predictor association has been established, they fire

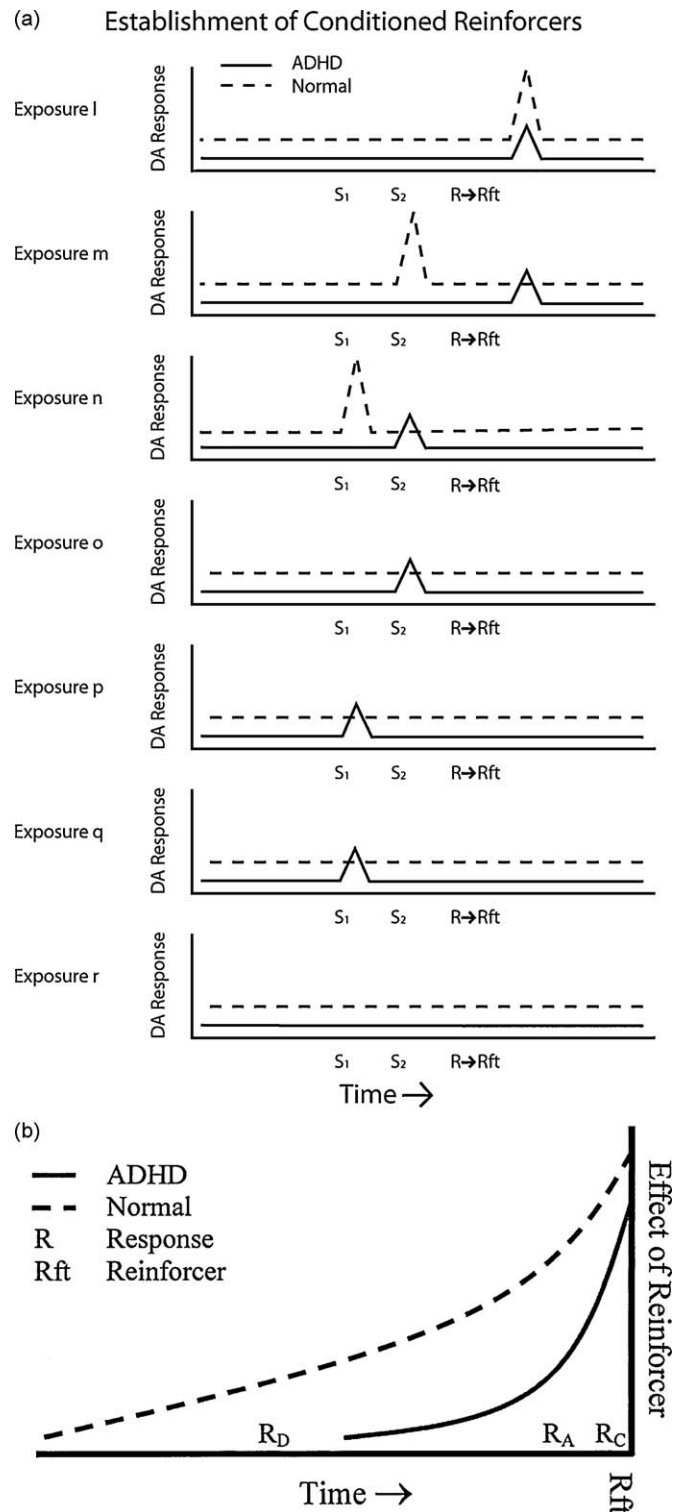


Fig. 1. Establishment of conditioned reinforcers (a) and the delay-of-reinforcement gradient (b) according to DDT (reprinted with permission from Sagvolden et al., 2005). (a) Normally, following a reinforcer (Rft), there is a short-lasting dopamine cell firing. This dopamine activity level is gradually transferred to the earliest stimulus predicting future reinforcers. When the relation between behavior and reinforcers is established, and thus there is a stable stimulus (S1, S2)–response (R)–reinforcer (Rft) relationship, there is no change in dopamine activity. DDT predicts that in ADHD, hypofunctioning dopamine systems slow this process. (b) Theoretical delay-of-reinforcement gradients. The effect of a reinforcer is more potent when the delay between the response and the reinforcer is short than when the delay is long. The delay gradient is predicted to be steeper and shorter in children with ADHD than in typically developing children.

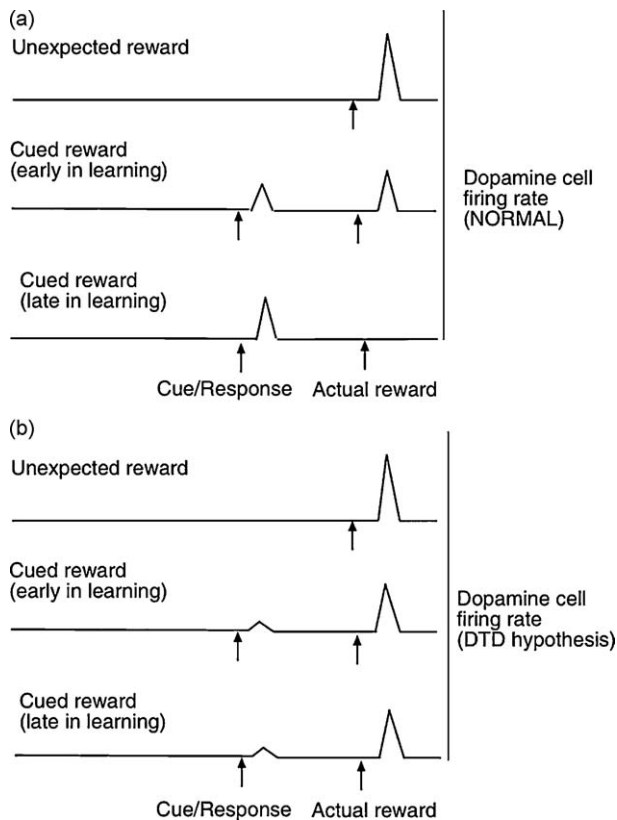


Fig. 2. Establishment of conditioned reinforcers in healthy individuals and in ADHD according to DTD (reprinted with permission from Tripp and Wickens, 2008). (a) Normal transfer of dopamine cell firing. Unexpected reinforcement is a potent stimulus for dopamine cell firing activity. Early in learning, dopamine cell firing responses transfer to cues that predict later reinforcers. They may also transfer to responses, which can act as cues that predict reinforcers. Later in learning, responses to cues may dominate over responses to actual reinforcers. (b) Dopamine cell firing in individuals with ADHD according to DTD theory. Dopamine cell firing fails to transfer to the cues that predict later positive reinforcers.

only to its predictor. The DTD theory proposes that this anticipatory DA cell firing is disturbed in ADHD. Accordingly, children with ADHD should evidence problems with the anticipation of rewards and poorer control of behavior. In addition, less effective learning would be expected with delayed compared to immediate rewards, as well as under conditions of partial compared to continuous reinforcement. The weaker conditioning would result in faster extinction of behavior, resulting in a weaker influence of reinforcers on behavior over longer periods of time. Dopamine firing in response to actual rewards is assumed to be normal.

In summary, both theories suggest that individuals with ADHD would prefer immediate over delayed rewards (see prediction 1 in Table 1), need frequent reinforcement to learn optimally (prediction 3), show impaired learning in response to reinforcement (prediction 4), and show impaired integration of earlier experiences of reinforcement (prediction 5). However, the underlying neurobiological causes of these behaviors differ between the models. The DDT proposes that a lower level of tonic DA in the fronto-limbic circuitry in the brain (prediction 10) is responsible for a smaller phasic DA signal to rewards (prediction 11), a slower shift in the DA response (prediction 12) and smaller DA response to reward cues (prediction 13). According to DDT, the hypodopaminergic state would result in a blunted dip in phasic DA to the omission of reward, resulting in slower extinction of behavior (prediction 14). The DTD, on the other hand, predicts normal levels of tonic DA and normal DA responses to actual

rewards, but a failure of transfer of the DA signal and reduced DA responses to reward cues, possibly related to an inability to suppress the firing of DA cells to an unconditioned stimulus (see Tripp and Wickens, 2009). They also suggest faster behavioral extinction and increased sensitivity to changes in reinforcement contingencies (prediction 6), because of an undue influence of unexpected (new) rewards.

2.2. Brain-pathway deficit models

The dual pathway model (Sonuga-Barke, 2002, 2003) and the integrative theory of ADHD (Nigg and Casey, 2005) suggest abnormalities in reinforcement sensitivity in ADHD through a deficiency in the activation of prefrontal pathways. The dual pathway model proposes that there are at least two independent neural circuitries related to ADHD, namely ventrolateral and dorsolateral cortico-striatal circuitry subserving executive processes, and mesolimbic (medial-prefrontal and orbitofrontal) ventral striatal circuitry subserving motivational processes. Both pathways are driven by DA activity. Specifically, Sonuga-Barke proposes that ADHD is associated with delay aversion, resulting in relatively strong preferences for smaller immediate rewards over larger delayed rewards, due to abnormalities in the motivational pathway. According to this model escape from delay is the key reinforcer for children with ADHD, as delay appears to be associated with a negative valence in ADHD (see Plichta et al., 2009). When delay cannot be reduced, children will engage in behaviors that reduce the perception of the length of delay, or serve as immediate reinforcers, such as fidgeting and attending to alternative stimuli (e.g., distractibility). Later versions of the model (Sonuga-Barke, 2003) suggest that the motivational pathway is associated with aversion to pre-reward delays rather than delay in general. In terms of the neurobiological mechanisms responsible for delay aversion it is proposed that in ADHD there is reduced efficiency of DA in reward circuits signaling future rewards, acknowledging early development of DDT concerning a steeper and shorter delay-of-reinforcement gradient in children with ADHD (Sagvolden et al., 2005).

The integrative theory of ADHD (Nigg and Casey, 2005) suggests three dysfunctional neural circuits in ADHD that are implicated in the support of learning 'what' may occur (fronto-striatal circuit) and 'when' it may occur (fronto-cerebellar circuit), as well as a system that is involved in evaluating the 'emotional significance' of events (prefrontal cortex). Emotional information that is irrelevant or violates current goals is suggested to initiate top-down prefrontal cortex regulation of the subcortical structures. Signals from the ventral striatum to the prefrontal cortex would enhance the tendency to approach positive stimuli, while signals from the ventral amygdala would invoke avoidance behavior. Therefore, problems in ADHD in the 'when' expectation of positive and negative stimuli, such as reward and non-reward, may result in aberrant approach and avoidance behaviors. Children with ADHD are suggested to inadequately initiate the top-down prefrontal structures and therefore, fail to show the demanded and expected behavior. Once initiated, however, the prefrontal structures are intact in activating the expected behavior.

In summary, both brain-pathway deficit models predict that children with ADHD prefer immediate over delayed rewards (prediction 1). The dual pathway model suggests this to be caused by an aversion to delay, rather than a reduced predictability of reinforcement (integrative theory). In contrast to DDT or DTD, the integrative theory explains reinforcement-learning deficits in ADHD (prediction 4 and 6) by inadequate initiation of prefrontal structures through the inability to detect irregularities in reinforcement patterns. When explaining the neurobiology of ADHD the neurochemical deficit models differ from the

Table 1

Predictions on altered reinforcement sensitivity in ADHD.

Predictions	Models							
	DDT (Sagvolden et al., 2005)	DTD (Tripp and Wickens, 2008)	Response modulation theory (Patterson and Newman, 1993)	Go/No-Go learning model (Frank, 2005)	Extended temporal difference model (Williams and Dayan, 2005)	Integrative theory (Nigg and Casey, 2005)	Dual pathway model (Sonuga-Barke, 2003)	Empirical evidence
Behaviorally testable								
1. Stronger discount of future rewards over immediate rewards	V	V	–	V	V	V	V	Evidence ^{a,e}
2. Stronger discount of future rewards decreases with practice (more trials)/reduced by environmental predictability	V	–	–	–	V	V	V When this leads to delay reduction	More research needed
3. Poorer performance under partial or discontinuous reinforcement schedules/normal performance under continuous reinforcement schedules	V	V	–	V	–	–	V	Some evidence ^{a,e} , more research needed
4. Impaired reinforcement-learning and acquisition of behavior	V	V	–	V	V	V Due to a predictability failure	–	Some evidence ^a
5. Impaired integration of earlier reinforcers, undue influence of individual instances of reinforcement	V	V	–	V	V	–	–	Some evidence ^a
6. Impaired ability to change behavior in response to changes in reinforcement contingencies	V	X Increased ability	V	V	–	V	–	More research needed
7. Impaired response to conditioned than to actual reinforcement	X Impaired response to both	V	V	V	V	V Due to a predictability failure	–	More research needed
8. Problems with adding new contingency information in working memory	V	X Updating too fast	–	V	–	V	–	More research needed
9. Behavioral inhibition less under the influence of cues of aversive stimuli, particularly when reward is available	–	–	V	X No-go learning independent of go learning	–	V Impaired detection of emotional significance of aversive cues	–	No evidence ^{a,b} , more research needed
Neurobiologically testable								
10. Lower level of tonic dopamine in the fronto-limbic circuitry of the brain (hypodopaminergic state)	V	X Normal levels	–	V	V Only for a specific cohort	V	V	Some evidence ^{c,d}
11. Smaller phasic dopamine response to actual reward	V	X Normal response	–	V	–	–	X Normal response	Determine how
12. Slower shift in dopamine from actual reward to reward cue	V	X Speed of shift is normal	–	–	V	V	V	to test this
13. Reduced phasic anticipatory dopamine release in striatum to reward cues (MPH facilitates this release)	V (V)	V (V)	–	V (V)	V (–)	V (–)	V (V)	Determine how to test this
14. Slower rate of extinction (MPH normalizes extinction) → This prediction is also testable behaviorally	V (speeds up with MPH)	X Faster extinction (slows down with MPH)	–	–	–	–	–	Some evidence ^b (studies on MPH needed)
15. Striatal problems are at the core of impaired activation of the frontal cortex	V	V	V	V	V But, more processes impaired	V But, more pathways impaired	V But, more pathways impaired	Some evidence ^a (studies on MPH needed)

Note: V = agree; X = disagree; – = not addressed. MPH = methylphenidate. The predictions 1, 3, 5, 7, 11, 13, and 14 are taken directly from Tripp and Wickens (2007), page 8 (this paper also discusses the DDT and the TD Learning model of Williams and Dayan). Predictions 2, 10 and 12 are directly derived from the model of Williams and Dayan (2005), page 174. Predictions 6 and 15 are derived from Nigg and Casey (2005), page 798. Predictions 4 and 8 are derived from Frank et al. (2007a,b), page 1674. Prediction 9 is derived from Patterson and Newman (1993). Table 1 shows that predictions 1–5, 13 and 15 are shared (thus agreed upon by more than 1 model, while predictions 6–12 and 14 are not agreed upon by all models).

^a Evidence from behavioral studies.

^b Evidence from fMRI or ERP studies.

^c Evidence from PET studies.

^d Evidence from pharmacological studies.

^e Evidence from animal-model studies.

brain-pathway deficit models in the level of analyses. While the brain-pathway models focus on the ‘higher-level’ processes, such as interactions between brain structures, DDT and DTD focus on ‘lower-level’ processes such as DA functioning of the striatum.

2.3. Neurocomputational models

The extended temporal difference model of Williams and Dayan (2005) proposes, like DTD, that ADHD is associated with inadequate transfer of DA signaling from reward delivery to reliable predictors. This model attempts to explain impulsive behavior in ADHD, as measured by impaired performance on the delayed response time tasks involving the choice between a small immediate and a large delayed reward. In developing their model Williams and Dayan (2005) assumed that performance on a delayed response task is dependent on four parameters that are derived from temporal difference learning: brittleness (the extent to which behavior is based on newly learned responses or on existing knowledge), action bias (a measure of the model's preference for action over inaction), learning rate (the rate at which behavior changes, associated with heritable factors that influence an enhanced or suppressed release of DA, Schultz et al., 1997), and discount factor (decrease in value of reward in the future as compared to immediate reward). The model demonstrates that all four parameters may influence the preference for immediate rewards. Importantly, both over- and under-regulated behavior (brittleness) and hypo- and hyperfunctioning of the DA signal to rewards (learning rate) could explain impulsivity.

The neurocomputational Go/No-Go learning model (Frank, 2005; Frank et al., 2007b) suggests that ADHD is associated with low striatal DA which can account for both motivational and working memory deficits, while an independent, likely noradrenergic, mechanism may account for variability in response latencies and poor response inhibition. This computational model simulates dynamic DA interactions within circuits linking the basal ganglia with the frontal cortex. The model suggests that phasic DA signals that occur during the presence of reinforcement (e.g., Schultz, 2002) modulate the ‘gating’ system in the striatum, which may activate or inhibit the execution of actions represented in the frontal cortex (see Fig. 3). In particular, learning within this gating system facilitates the selection of rewarding actions and prevents selection of those that are less rewarding (Frank, 2005). If striatal DA is reduced in

ADHD, the model predicts ADHD-related deficits in action selection, particularly during positive reinforcement-learning. The model predicts a decrease in go-signals for appropriate motor behavior and a raised threshold to updating task-relevant information into working memory, in particular once the actual stimulus is no longer present or in the face of distracters (Frank et al., 2007b). In addition, DA reductions in ADHD would lead to impaired representation of cues that predict future rewards.

These computational models agree with the other models on several behavioral predictions (prediction 1, 3–8), however, they differ in the suggested underlying mechanisms of these behaviors. According to the extended temporal difference model, several aspects of behavior (rather than a single mechanism) relate to impairments in reinforcement sensitivity in ADHD. For example, steeper delay discounting in ADHD (prediction 1) is dependent on factors such as enhanced or suppressed DA responses to rewards, but also the system's preference for action over inaction, or predictability of the reward (prediction 2). The Go/No-Go learning model suggest inadequate updating of working memory (prediction 8) to be an important aspect of impairments in learning from reinforcement (prediction 6). According to this model, a hypodopaminergic state (prediction 10) is responsible for blunted DA responses to reward (prediction 11) and impaired positive reinforcement-learning (prediction 6). The extended temporal difference model, on the other hand, suggests that both a hypo- or hyperdopaminergic state may be responsible for altered reinforcement sensitivity in ADHD.

2.4. Descriptive model of sensitivity to punishment

In contrast to the models described above, the response modulation theory (Newman and Wallace, 1993; Patterson and Newman, 1993) focuses on dysregulation of activity in the sympathetic nervous system. Disinhibited behavior such as that observed in ADHD is suggested to be a consequence of impaired passive avoidance behavior, which is the ability to inhibit a motivated response when the possibility of punishment exists. Impaired passive avoidance is suggested to result from over reactivity of the sympathetic nervous system to positive stimuli, such as reward, that overrules a sympathetic response to aversive stimuli. This model is partly based on the work of Gray (1982, 1987) who postulated that behavior is modulated by two separate, but interacting brain systems: The behavioral activation system (BAS) which initiates approach behavior, and the behavioral inhibition system (BIS) which initiates extinction behavior or passive avoidance. Rewards or non-punishment activates the BAS, while aversive stimuli or non-rewards activate the BIS. According to the response modulation theory impulsive individuals experience increased activity of the BAS, which results in an active search for rewards and a lack of passive coping with aversive stimuli. Passive coping with aversive stimuli includes the coupling of the aversive stimulus to the preceding behavioral response in order to avoid punishment in the future (response modulation). According to Gray the BAS is dominated by the mesolimbic DA pathway, including the ventral tegmental area and ventral striatum. It is not specified if increased BAS activity reflects overactive or underactive DA activity within this system.

The main prediction of this model is that ADHD is associated with an inability to inhibit initiated behavior in response to aversive cues (prediction 9), particularly when aversive cues are presented simultaneously with reward.

2.5. Comparison of the main features of the models

When testing the predictions of these models, or when explaining behavioral findings using these models, it is important

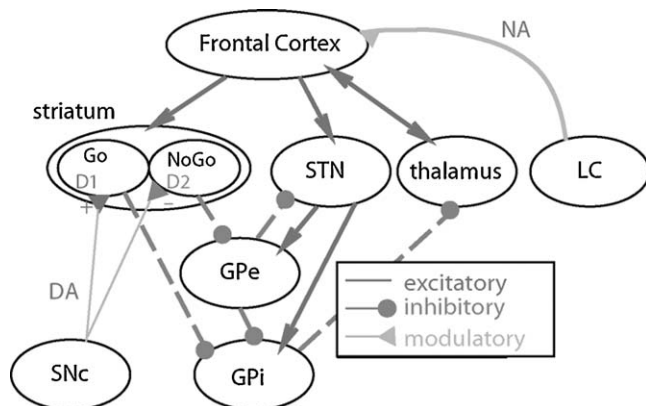


Fig. 3. Parameters of the Go/No-Go Learning model (reprinted with permission from Frank, 2005). Note: GPI = internal segment of globus pallidus; GPe = external segment of globus pallidus; SNc = substantia nigra pars compacta; STN = subthalamic nucleus. Striato-cortical loops including the direct ('Go') and indirect ('No-Go') pathways of the basal ganglia. The Go cells disinhibit the thalamus via GPI, facilitating the execution of an action represented in cortex. The No-Go cells have an opposing effect by increasing inhibition of the thalamus and suppressing action execution. Dopamine from the SNc excites synaptically driven Go activity via D1 receptors and inhibits No-Go activity via D2 receptors.

to acknowledge the between-model differences in the level of explanation offered. When explaining reinforcement sensitivity in ADHD, some models are mainly neurobiological in nature (DDT, DTD, Go/No-Go learning model), while other models are more behavioral, for example explaining delay aversion (dual pathway model, extended temporal difference model). Models differ in terms of explaining single versus multiple deficits. For example, DDT and DTD attempt to explain a single core deficit in terms of lower-level reinforcement systems, such as the DA transfer back in time. Other models attempt to explain multiple deficits in terms of higher-level systems, such as bottom-up activation of prefrontal systems (e.g., integrative theory). The 'lower-level' models focus particularly on the midbrain DA system (DDT, DTD, response modulation theory) or basal ganglia and associated routes (Go/No-Go learning model). The 'higher-level' models describe cortical-subcortical pathways such as fronto-striatal/fronto-cerebellar routes (integrative theory) or fronto-striatal/fronto-limbic routes (dual pathway model). The extended temporal difference model, on the other hand, focuses on general aspects of the brain's flexibility and sensitivity. Another difference between the models is that some have a particular focus on reinforcement-learning aspects (DDT, DTD, dual pathway model, Go/No-Go learning model). Finally, while the dual pathway model and the integrative theory describe reinforcement sensitivity problems in those with symptoms of hyperactivity/impulsivity and inattention, other models focus specifically on explaining hyperactive/impulsive problems (DDT, response modulation theory).

3. Research methods

In order to determine which of the proposed underlying mechanisms are responsible for altered reinforcement sensitivity in ADHD, we need to carry out studies that focus on multiple causes and mechanisms (or focus on a single cause with multiple outcomes). What is required is a combination of technologies and/or research methods that investigate these mechanisms, rather than focusing on outcomes only. This is not an easy job, since testing the neurobiological mechanisms will be particularly challenging, as current research methods do not permit examination of all predictions directly in humans. For example, the prediction that levels of extracellular DA in the ventral striatum (prediction 10) are low in individuals with ADHD (e.g., DDT) is currently difficult to test in humans.

A 'multi-level' research approach calls for studies that combine performance measures with intermediate measures between behavior and neurobiology such as neural imaging or psychophysiological measurements. Recent pharmacological MRI research shows that DA-release is correlated with BOLD response in ventral striatum (see for a review Knutson and Gibbs, 2007). In addition, DA activity to positive reward prediction errors is reflected in ERP signals measured at the cortex (Holroyd et al., 2008). And although the number of studies using neural imaging is scarce (Plichta et al., 2009; Rubia et al., 2009; Scheres et al., 2007; Ströhle et al., 2008; van Meel et al., 2005), the results are encouraging and provide a basis for further research into the brain reward circuitry and its role in ADHD. Another method to test the proposed mechanisms responsible for ADHD is the use of pharmacological interventions that affect the catecholamine system. Stimulants such as methylphenidate (MPH) preferentially block the reuptake of DA in the striatum (Schiffer et al., 2006; Seeman and Madras, 2002; Volkow et al., 2001) and may enhance the saliency of positive stimuli, such as rewards (Volkow et al., 2004). Additionally, atomoxetine blocks the reuptake of norepinephrine in the frontal cortex, which increases the concentration of norepinephrine as well as DA in the frontal cortex (Bymaster et al., 2002). Currently, there are only a few studies on the impact of stimulants on reward

sensitivity in ADHD, focussing mainly on MPH (Pelham et al., 1986; Solanto et al., 1997; Tripp and Alsop, 1999).

Many of the neurobiological predictions can be tested using animal models of ADHD, although it should be kept in mind that animal models only approach aspects of a human disorder, rather than being equivalent (Alsop, 2007; Russell, 2007). Tasks that manipulate reinforcement frequency that distinguish the most widely accepted animal model for ADHD, the spontaneously hypertensive rat (SHR) from control rats, have been tested in humans with similar findings (see Aase and Sagvolden, 2006; Sutherland et al., 2009). Finally, an increasing number of studies demonstrate an association between the development of ADHD symptoms and genes that influence the dopamine and serotonin system. More specifically, ADHD symptoms have been associated with DA transporter genes (e.g., DAT1), DA receptor genes (e.g., DRD4 and DRD5), as well as serotonin transporter genes (e.g., 5HTT) and serotonin receptor genes (e.g., HTR1B) (see Gizer et al., 2009 for a meta-analysis). Although absent so far, studies examining the link between these genes and abnormalities in reinforcement sensitivity in ADHD could increase our understanding of the underlying (neural) mechanisms.

Finally, in addition to encouraging researchers to use different methodological approaches, what is needed is crosstalk between the fields such as neurobiology, neuroanatomy and computer science. Researchers who work with individuals with ADHD typically do not have the facilities or the skills to study predictions at the level of basic neuroscience. Different disciplines can benefit from shared knowledge on neural mechanisms of DA functioning. For example, Reynolds et al. (2001) showed that stimulation of the substantia nigra increases the strength of the DA-dependent cortico-striatal connectivity in order to increase learning of behavioral responses (see also Kerr and Wickens, 2001). This indicates that DA increases the efficiency of signal transfer in the fronto-striatal pathway that is necessary for the acquisition of behavioral responses.

4. Predictions and available evidence

Table 1 provides an overview of the behavioral and neurobiological predictions from the seven models. It is clear from Table 1 that some of the predictions are shared (agreed upon by more than one model), while others are unique or contradictory (not shared by all models). Where possible, we have indicated whether the various models agree with, disagree with, or do not address each prediction. By comparing predictions with the experimental evidence we can identify: if and how the predictions can be tested, which predictions are important to test, and which predictions are supported by existing experimental evidence. The behaviorally testable predictions are considered first (predictions 1–9), followed by the neurobiologically testable predictions (predictions 10–15).

4.1. Behavioral predictions and available evidence

4.1.1. Delay discounting (prediction 1 and 2)

Behavioral studies (see Sonuga-Barke et al., 2008 for review) indicate that children with ADHD show relatively strong preferences for smaller immediate over larger delayed rewards, thus demonstrating greater reward delay discounting. The extended temporal difference model suggests that delay discounting is caused by learning factors, but also factors from the environment (e.g., context) (Williams, 2008). Indeed, various important contextual factors, such as the length of delay, number of trials or amount of practice (prediction 2) are predicted to modulate these effects, although these factors are rarely studied (but see Barkley et al., 2001; Scheres et al., 2006, in press). The results from

Barkley et al. and Scheres et al. indicate that the preference for immediate rewards in ADHD may diminish when using varied versus fixed delay durations or when the magnitude of the delayed reward is relatively large.

Studies with the SHR rat have shown that when rewards were delivered after a delay, response acquisition of the SHR was relatively slow and response rates were relatively low. With immediate rewards, on the other hand, the SHR exhibited higher response rates than control strains (e.g., Hand et al., 2006; Johansen et al., 2005; Sutherland et al., 2009). This is consistent with prediction 1.

4.1.2. Partial reward (prediction 3)

Up to now, only a few studies have compared the effects of continuous and partial reinforcement on task performance (see Luman et al., 2005; Douglas and Parry, 1994). The results provide some evidence for impaired performance during intermittent versus continuous reward delivery in ADHD. Some of the studies provided continuous feedback during the partial reinforcement schedule, which may have confounded the results (Douglas and Parry, 1994). Studies that actually manipulate reinforcement frequency are rare, but do document increased variability of responding under intermittent reinforcement conditions in ADHD (Aase and Sagvolden, 2006). In addition, there is evidence of impaired reward-based learning in ADHD when rewards are not contingent, but probabilistic, and thus infrequent (Frank et al., 2007a). At the same time, learning from contingent rewards also seems to be impaired in children with ADHD (Luman et al., 2009).

Studies with the SHR show that response rates differ from control rats when rewards are delivered infrequently, while no differences are observed with high (frequent) rates of reward (Sagvolden et al., 1993), offering some support for prediction 3.

4.1.3. Reinforcement-learning (predictions 4–7)

All models that included a learning component, except the response modulation theory and dual pathway model, predict deficient reinforcement-learning in ADHD (prediction 4). The impact of reward on learning curves in individuals with ADHD has been studied twice (Frank et al., 2007a; Luman et al., 2009). Both studies indicate that 'go-learning' (learning from positive feedback) is impaired in ADHD, providing support for prediction 4. Prediction 5 focuses on difficulties incorporating past experiences of reinforcement. To date, only 1 study has investigated this issue (Tripp and Alsop, 1999), providing support for prediction 5. The models differ with respect to prediction 6. While DDT, Go/No-Go learning model and response modulation theory predict slower changes in behavior in response to changes in reinforcement contingencies, DTD predicts that individuals with ADHD are able, like controls, to adapt their behavior in response to actual rewards. To the best of our knowledge, this has not been tested experimentally. All models except DDT and dual pathway model predict that learning from reward cues is disturbed in individuals with ADHD (prediction 7), while the response to actual reward is normal. This prediction has not yet been addressed in behavioral studies.

4.1.4. Updating working memory (prediction 8)

Some models (e.g., Go/No-Go learning model) suggest that altered sensitivity to reinforcement in ADHD is related to impaired updating of reinforcement information in working memory. According to DTD, impaired integration of previous reinforcement experiences (see prediction 5) may be caused by an increased updating of reinforcement information in working memory (although this was not the primary focus of the model). This prediction suggests that reinforcement sensitivity may be second-

ary to cognitive, rather than motivational deficits in ADHD (see for review Johansen et al., 2009), but this has not been studied so far.

4.1.5. Reward omission and aversive stimuli (predictions 9 and 14)

Both the response modulation theory, Go/No-Go learning model and integrative theory suggest an impaired behavioral reaction to aversive stimuli such as punishment in ADHD. So far, the majority of studies on ADHD focus on positive reinforcement (stimuli that increase the likelihood of behavior), with only a few considering aversive stimuli such as response cost (removing rewards when responses are incorrect; see Gomez, 2003; Luman et al., 2005, 2007; van Meel et al., 2005). Event-related potential studies in ADHD suggest that brain activity 500 ms following response cost is larger in ADHD than controls (van Meel et al., 2005). This may be related to a reduced expectancy of negative outcomes in ADHD, an explanation that would be in line with the integrative theory. These studies do not support prediction 9, although the impact of response cost may be distinct from that of punishment (stimuli that reduce the frequency of responses, such as a loud noise). The Go/No-Go learning model suggests that learning from aversive stimuli is independent of learning from rewards, although this has not been tested so far.

Finally, it will be important to study the impact of reward omission on extinction in ADHD (prediction 14). According to DDT, children with ADHD will show impaired (slower) extinction, while the DTD and the extended temporal difference model suggest enhanced (faster) extinction of behavior in ADHD. This has been studied only once using a behavioral task and the results of this study suggests slower extinction in ADHD (Itami and Uno, 2002).

In sum, from amongst these behaviorally testable predictions, there is initial evidence that ADHD is related to: stronger discounting of larger future rewards over smaller immediate rewards (prediction 1; e.g., Johansen et al., 2005; Sonuga-Barke et al., 2008), poorer performance under partial compared to continuous reward schedules (prediction 3; e.g., Luman et al., 2005; Sagvolden et al., 1993), impaired reinforcement-learning and acquisition of behavior (prediction 4: Frank et al., 2007a; Luman et al., 2009), and impaired integration of earlier reinforcers (prediction 5: Tripp and Alsop, 1999). There is currently no evidence for reduced sensitivity to aversive stimuli in ADHD (Prediction 9; e.g., Luman et al., 2005; van Meel et al., 2005). More research is needed to test the other predictions (2, 6–8).

4.2. Neurobiological predictions and available evidence

4.2.1. Levels of tonic dopamine (prediction 10)

Positron emission topography (PET) studies show mixed evidence of abnormal striatal DA transporters density in individuals with ADHD that is indicative of DA availability. Three studies have shown abnormally high levels of DA transporter density in ADHD (Dougherty et al., 1999; Krause et al., 2000; Spencer et al., 2005), while a recent study suggests unusually low density of DA transporters in this group (Volkow et al., 2009).

Studies with MPH (assuming that MPH increases DA availability) show evidence of normalized reinforcement sensitivity as measured by increases in performance and learning abilities under medication compared to placebo (Frank et al., 2007a; Pelham et al., 1986; Solanto et al., 1997; Tripp and Alsop, 1999; Wilkison et al., 1995), possibly supporting prediction 10. However, more research is needed to understand the mechanism of action of MPH.

4.2.2. Dopamine responses to reward (predictions 11, 12 and 13)

Currently, no studies have tested if the DA response to rewards is reduced (DDT, Go/No-Go learning model) or normal (DTD, dual pathway model), if the transfer of DA cell activity to reward cues is slowed (DDT, extended temporal difference model, integrative

theory, dual pathway model) or fails to occur (DTD). Recent fMRI studies suggest attenuated anticipatory BOLD responses to reward in adolescents and adults with ADHD (Plichta et al., 2009; Scheres et al., 2007; Ströhle et al., 2008), suggesting reduced DA responses to anticipated rewards (prediction 13). This is supported by ERP studies that suggest smaller event-related brain activation to (anticipated) rewards in children with ADHD compared to typically developing controls (Holroyd et al., 2008; van Meel et al., 2007).

4.2.3. Impaired activation of the prefrontal cortex (prediction 15)

This prediction has not been tested so far, but can possibly be tested using diffusion tensor imaging (DTI) in humans.

In sum, from amongst these neurobiologically testable predictions, there is mixed evidence that ADHD is related to a lower level of tonic DA in the fronto-limbic circuitry (prediction 10; e.g., Dougherty et al., 1999; Volkow et al., 2009), and some evidence of an attenuated DA response to anticipated rewards (prediction 13; e.g., Scheres et al., 2007; van Meel et al., 2007). More research is needed to test the other neurobiological predictions (11–12, 14, 15).

5. Moderating factors and related issues

When testing the predictions using ‘multi-level’ methods, researchers need to be aware of several confounding issues that may impact reinforcement sensitivity in ADHD.

5.1. Group-related moderators

Factors such as ADHD subtypes, comorbid symptoms, gender, and age may act as potential moderators of reinforcement sensitivity in ADHD. There is some evidence that suggests that certain alterations in reward sensitivity are specifically associated with symptoms of hyperactivity/impulsivity (such as steep temporal discounting and reduced activation of the ventral striatum; Scheres et al., 2007, 2008, *in press*) while others may be associated with symptoms of inattention (sensitivity to penalty, Huang-Pollock et al., 2007). Since ADHD is a very heterogeneous disorder at the symptom level, it is possible that alterations in reinforcement sensitivity are a distinct, neurobiologically informed subtype of ADHD (as suggested by the dual pathway model), or are an overlapping, but independent condition (e.g., Tripp and Wickens, 2008). In addition, researchers should consider the specificity of altered reinforcement sensitivity in ADHD. ADHD is comorbid with several psychiatric conditions including conduct disorder, depression, anxiety, and substance abuse, all of which have been linked to reinforcement dysfunctions (e.g., Forbes et al., 2007; Newman and Wallace, 1993; Raine, 1993). Neuroanatomical data suggests that children with ADHD show a delay in cortical maturation (e.g., Shaw et al., 2007) and therefore certain ADHD-related alterations in reinforcement sensitivity may dynamically change over time (but see Scheres et al., 2007; Ströhle et al., 2008). Furthermore, animal research suggests that chronic use of psychostimulants such as MPH alters the DA system and reward sensitivity in adulthood, even long after discontinuation (see Grund et al., 2006 for review). Thus, research on the impact of a long-term history of medication use on altered reward sensitivity and the workings of the DA system is needed.

5.2. Reinforcement-related moderators

Altered sensitivity to reinforcement may not be restricted to tangible rewards such as tokens that can be exchanged for a gift. The impact of verbal praise or a smiling face (or the lack thereof) may differ between children with ADHD and typically developing

peers (Geurts et al., 2008; Kohls et al., 2009). Therefore, the impact of a range of reward types needs to be studied. Finally, although it is a useful start to study the predictions using Table 1, studying motivational processes in isolation has limited ecological validity. It is becoming clear that children with ADHD suffer from motivational as well as cognitive problems, which may act as interacting phenotypes (see Johansen et al., 2009 for a review). Although there is already a number of studies examining the positive effects of reward on cognitive performance, some cognitive functions may be more sensitive to motivational manipulations than others (see Luman et al., 2005 for a review). Research is needed that examines how reward sensitivity may further impair cognitive functioning in ADHD, and studies the neural basis of this interaction between reward sensitivity and cognitive performance. Finally, research is needed to examine the influence of environmental factors on reward sensitivity in ADHD. For example, children with ADHD may act differently to reinforcers in a new environment (e.g., laboratory setting) as compared to a well-known environment (e.g., classroom setting).

6. Conclusion and research agenda

The goal of the current paper was to provide an updated review of seven neurobiologically meaningful models on reinforcement sensitivity in ADHD, to identify specific predictions and research methods required to test these predictions, and review the available experimental evidence. Since DA responses to reward are essential for learning processes (e.g., Frank, 2005; Tripp and Wickens, 2008) and motivation (Volkow et al., 2001), insight into reward sensitivity in ADHD can increase the efficacy of behavioral treatment programs that use reinforcers to shape behavior and increase academic performance. The evidence indicates that only 8 of the 15 predictions (1, 3–5, 9, 10, 13, 14) have been tested so far, in part because for some predictions (11, 12, 15) the best method of evaluation still needs to be determined. Most predictions that have been tested, however, are agreed upon by most models (1, 3–5, 13), making it difficult to discriminate between the different models. To date, many studies have focused on behavioral changes in the face of reward and response cost over non-reward, and on delay discounting (see Luman et al., 2005). These are behavioral phenomena that are predicted by most models. There is also some evidence for more ‘unique’ predictions. For example, it has been shown that, like controls, children with ADHD are sensitive to aversive stimuli (prediction 9) arguing against the response modulation theory (Patterson and Newman, 1993). However, more research on the impact of punishment in ADHD is needed. Also, if replicated, slower extinction in ADHD (prediction 14) would argue against DTD (Tripp and Wickens, 2008).

Moreover, studies need to explain underlying cognitive mechanisms, such as working memory problems (e.g., Frank, 2005) or the ability to detect irregularities within the environment (e.g., Nigg and Casey, 2005). Additionally, studies need to focus on neural mechanisms such as the midbrain DA system (e.g., Sagvolden et al., 2005), the basal ganglia and associated routes (Frank, 2005), and cortical-subcortical interactions (e.g., Nigg and Casey, 2005). By testing ‘unique’ neurobiological predictions, we can determine what mechanisms underlie altered reinforcement sensitivity in ADHD. For example, the DTD differs from other studies in suggesting intact DA responses to rewards (prediction 11) as compared to reward cues (prediction 13), resulting in impaired integration of earlier reinforcers (prediction 5). Another important future goal is to specify which subgroups of children with ADHD show altered reinforcement sensitivity (e.g., Sonuga-Barke, 2003), for example, by looking at ADHD subtypes or comorbidities. Finally, future studies should take into account the many confounding factors that could influence reinforcement

sensitivity in ADHD, such as medication history, reinforcement type, and environmental factors.

7. Limitations

First, this review focuses on the role of DA in reinforcement mechanisms, while other neurotransmitter systems, such as serotonin and norepinephrine (e.g., Blum et al., 2000) are not considered. Since these systems are described less extensively in relation to ADHD, they were excluded from the current review. Second, models that are not grounded in neuroscience are excluded, although these models have been very influential in explaining behavioral aspects of ADHD (e.g., Douglas, 1989; Haenlein and Caul, 1987; Sergeant et al., 1999). These earlier models of altered reinforcement sensitivity in ADHD may be updated with a neurobiological explanation of their behavioral predictions (see Luman et al., 2005 for a review). Third, emotional responses to altered reinforcement sensitivity, such as negative emotions during reward delay in ADHD (Plichta et al., 2009), are not discussed. As we are only beginning to understand the neurobiological mechanisms of emotion processing in ADHD (e.g., altered amygdala activation during reward delays; Plichta et al., 2009), these studies fell outside the scope of this review. Fourth, although we have attempted to indicate as best as we can whether the models agree or disagree over the 15 predictions depicted in Table 1, the predictions are not conclusive and are sensitive to changes. Finally, it is argued above that studies are needed to test some of the neurobiological predictions. However, the current methods only allow us to test these mechanisms indirectly (e.g., by measuring BOLD responses or by using animal models). As a consequence, the importance of behavioral studies is emphasized in this review.

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